

PRADEEP A. DALVI

Principal Systems Architect | Distributed Systems & Kernel Specialist

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EXECUTIVE SUMMARY

Distinguished engineer with nearly **22 years of experience** specializing in high-performance distributed systems and networking protocols. Recognized expert in **Linux kernel development**, 10Gbps device drivers, and microservices architecture. Proven track record of bridging the gap between deep-level systems programming and executive-level strategy to build secure, resilient enterprise architectures.

TECHNICAL EXPERTISE & CORE INTERESTS

System Architecture & Distributed Computing

- **Scalable Systems:** Expert in architecting highly scalable, fault-tolerant microservices and distributed systems.
- **Distributed Stack:** Advanced proficiency in RabbitMQ, Apache HBase, Kafka, Apache Zookeeper, Apache Hive, and Apache Apex.
- **Architecture Principles:** Advanced practitioner of Object-Oriented Design (OOD) and Design Patterns to build modular, maintainable software.
- **Development Skills:** Comprehensive experience in high-level design, development, and system improvements.

Kernel & Systems Engineering

- **Linux Kernel:** Deep expertise in kernel-space development, Loadable Kernel Modules (LKM), and custom OS tuning.
- **Systems Programming:** Proficient in C/C++, Java, and Python with a focus on system software development.
- **Driver Development:** Proven track record of building complex network device drivers (4G/10G) from scratch & upstreaming to the mainline kernel.
- **Optimization:** Specialized in performance optimizations within the OS critical path.

Networking & Security

- **Network Infrastructure:** Expert in TCP/IP stack development, performance tuning, and TCP Offload Engine (TOE) for 10Gbps+ environments.
- **Security & Identity:** Extensive experience in Network Security (DPI, SSL Decryption, Sandboxing) and enterprise identity protocols including LDAPv3, Kerberos, RADIUS, and 2FA.
- **Advanced Debugging:** Strong debugging capacity for resolving complex, low-level system issues.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

PhonePe Ltd, Pune India

June 2017 – April 2026

Payments Infrastructure & Zero-to-One Launches

- **Founding Architect:** Established the Pune engineering hub as the first hire; built a PCI-compliant Payment Gateway from scratch, scaling to 1M+ daily transactions using MariaDB, RabbitMQ, and Aerospike.
- **Smart Routing Innovation:** Designed a proprietary Success Rate (SR) routing engine using a pattern-matching algorithm that achieved a 78% success rate, consistently outperforming internal ML models.
- **Strategic Integrations:** Led the deep-level Cybersource (Visa) integration, leveraging granular protocol control to drive a major Indian bank's success rate from below 50% to 70+%.
- **Ecosystem Expansion:** Pioneered the first-ever consumer-facing system outside the PhonePe app, architecting the web-based infrastructure to extend payment capabilities to the offline mass transit ecosystem.

Platform Scaling & Retention (Smaran - Reminders 2.0)

- **High-Throughput Distributed Systems:** Architected a multi-tenant, HBase-backed scheduling and delivery engine for 150M DAUs, scaling to 60M daily time-sensitive reminders.
- **Revenue & Growth Engine:** Drove 30% of total transactions across supported categories; achieved a 175% increase in Water Tax and 100% boost in Gas/Postpaid/Electricity reminders powered transactions.
- **Product Turnaround:** Rescued a stagnant product by optimising peak-hour stability, breaking a 11-month plateau to consistent ~10% MoM growth.
- **Operational Efficiency:** Engineered NAAM (Automated Accounting Sign-off), reducing manual overhead and accelerating new category onboarding by two weeks.

Reliability, Compliance & Transit

- **High Availability & BCP:** Engineered the first Active-Active Multi-DC Disaster Recovery (DR) for Recharge and BillPay, ensuring 99.99% uptime for mission-critical revenue streams.
- **Observability & Performance:** Leveraged deep architectural visibility to preempt outages ahead of automated monitoring during high-concurrency festive spikes and partner events.
- **Regulatory & Transit:** Spearheaded the NCMC (One Nation, One Card) issuance for Metro transit (1k+ issuances in week one) and managed rigorous PPI/RBI compliance.

Software Engineer

DataTorrent Software India Pvt Ltd · Pune, India

May 2015 – January 2017

- **Operator Development:** Engineered custom Apache Apex operators for projection, filtering, and laggard management in a real-time streaming.
- **AWS Cloud Integration:** Developed the integration suite for Amazon AWS, enabling seamless customer trials and cloud-native scaling.
- **Security Engineering:** Secured distributed communication within the RTS platform through advanced SSL/Security protocol implementation.

Principal Member of Technical Staff – I

Scalable Architecture Pvt Ltd · Pune, India

June 2014 – March 2015

Project in Brief: High-Performance Database Load Balancer.

- **Core Tech:** Kafka, Zookeeper, Python, CPU Isolation, Threading, Distributed Systems.
- **Performance:** Redesigned thread distribution for a High-Performance DB Load Balancer to optimise CPU utilisation.
- **Distributed Ops:** Implemented Kafka-based message queues for real-time data streaming and Zookeeper for cluster synchronisation.
- **Monitoring:** Built an asynchronous Python Alerting Engine for complex event processing across multi-node environments.

Senior Software Engineer

BlueCoat Systems (India) Pvt Ltd · Pune, India

October 2012 – May 2014

- **High-Performance Analytics:** Architected a high-speed web file extraction engine using Python, achieving significant throughput gains for large-scale security analytics platforms.
- **Threat Intelligence & Sandboxing:** Engineered a malware detection and reporting pipeline that automated the detonation of unknown files in a sandbox environment for real-time threat analysis.
- **Security Interoperability:** Developed remote notification integrations for enterprise SIEMs, including ArcSight and Syslog, ensuring seamless data flow across the security stack.
- **Network Visibility:** Designed and prototyped a specialized HAR (HTTP Archive) viewer to visualize and analyze high-speed web flows and performance bottlenecks.

Sr. Principal Development Engineer

Calsoft Pvt Ltd · Pune, India

January 2011 – September 2012

Services Company in Systems Software | Client: Solera Networks.

- **Linux Kernel Mainline:** Refactored memory management across 100+ network drivers to implement uniform buffer allocation, improving kernel network drivers.
- **Deep Packet Inspection (DPI):** Developed kernel-space PCAP parsing logic for proprietary high-speed forensic storage systems.
- **Security Engineering:** Prototyped ultra-high-speed SSL decryption and integrated multi-factor authentication (2FA/Kerberos/LDAP) into network security appliances.
- **Cloud & Virtualization:** Conducted technical prototyping for next-generation Virtualization and SaaS security products.

Independent Software Developer

Self-employed · Pune, India

November 2008 – January 2011

Explored Entrepreneurial Experience & Technical Leadership.

- **Director of Tech (eArth Solutions):** Led development of analytics engines for The Times of India & Hindustan Times.
- **Campaign Systems Architect:** Built mass-mailing infrastructure and global campaign sites for UN-nominated and local MP candidates (Arun Bhatia).
- **Consulting & Portals:** Executed technical assessments for iCrave.in and delivered SharePoint/OMS solutions for international and local clients (RGen Solutions, Café 1730).

Member of Technical Staff

Great Software Laboratory Pvt Ltd · Pune, India

June 2007 – November 2008

Product in Brief: Policy-based Application Level Firewall/Network Security Appliance.

- **Identity Management:** Achieved 10k auths/sec on MIPS64 Octeon processors via a custom multi-tier authentication interface for Cisco/Rohati.

System Software Engineer

LinSysSoft Technologies Pvt Ltd · Pune, India

January 2005 – April 2007

Services Company in Linux System Software | Client: Marvell Technology (formerly NetXen).

- **Driver Development:** Built 4G/10G Linux network device drivers from the ground up and managed the end-to-end linux kernel submission process.

EDUCATION

Bachelor of Engineering · Computer Engineering

University of Pune · India · 06/2004

First Class (Percentage: **61.40%**)